Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/11/2009
PI Name:	Lackner, James R. Ph.D.		
Project Title:	Predicting and Assessing Postural Reentry Disturbances		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	lackner@brandeis.edu	**Fax:**	FY 781-736-2031
PI Organization Type:	UNIVERSITY	**Phone:**	781-736-2033
Organization Name:	Brandeis University		
PI Address 1:	415 South Street		
PI Address 2:	Mail Stop: MS033		
PI Web Page:			
City:	Waltham	**State:**	MA
Zip Code:	02454-9110	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	07/01/2004	**End Date:**	09/30/2008
No. of Post Docs:	3	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	9
No. of Bachelor's Candidates:	9	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2008 per P. Denkins/JSC (5/07) NOTE: Funding reinstated to 6/30/2007, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bortolami , Simone (Brandeis University) DiZio, Paul (Brandeis University)		
Grant/Contract No.:	NNJ04HJ14G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project investigates a technique for predicting the severity of re-entry disturbances of posture and locomotion experienced by astronauts in space flight relevant environments and developing techniques for attenuating them. We are concerned not only with return to earth from orbital flight but also transfer to other force environments astronauts may encounter in the future, such as moon gravity, a rotating artificial gravity vehicle, or Mar's gravity. We have developed a hold and release (H&R) paradigm and mathematical model for quantifying multi-link postural performance, investigating the influence of flight-related factors and understanding neuromotor mechanisms . The linear model we have developed of upright posture as a four link inverted pendulum in a regime of small oscillations, a H&R experimental paradigm for producing transient postural oscillations, and data showing that the model fits the postural responses to H&R. The fitted parameters discriminate normal subjects and vestibular loss patients. This approach provides a way of quantifying astronauts' postural changes following re-entry, in terms of the latency of EMG responses, and characteristic stiffness and damping of the ankles, kness and hips. It may also provide a means of assessing the efficacy of countermeasures while astronauts are still in-flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The collective package of the H&R technique, the linear multi-link model and the analyses algorithms which this project showed were capable of discriminating postural stability in spaceflight analog environments could also be used easily by primary care doctors to assess the risk of falling, progress of posture-impairing conditions, and the effectiveness of treatment, while avoiding referring subjects to the current limited number of clinics running fatiguing and less sensitive tests. It would be possible to replace the laboratory system we used with a portable low-cost postural testing device, to facilitate widespread use for patients in the clinical or at home.
Task Progress:	Completed four major experiments during the project period, which are summarized below. One paper is published, one in submission and two more are in preparation. 1) Hold & Release in Parabolic flight. The aim of the presented study was to quantify recovery from simulated falls in altered force backgrounds and to investigate the mechanisms of recovery, using the H&R paradigm. We tested three healthy men and two healthy women ranging in age from 26 to 60. The experiment was conducted during the 0g and 1.8g phases of parabolic flight as well as during 1 g straight and level flight and 1 g on the ground pre- and post-flight. 0g trials were conducted with 50% body weight bungee cord loading. Subjects were tested both with their eyes open and closed in all force backgrounds. During testing, subjects stood feet side by side with the soles of their shoes anchored to a force plate attached to the deck of the aircraft. Loss of balance was elicited using the H&R paradigm. A horizontal force was applied to the subject's sternum, which he or she actively resisted. Removing the force caused subjects to sway forward, and they had to react to prevent falling. The exact time of release of the holding force was measured with a load cell. Motion of the head, shoulder, waist, knee and ankle were monitored with an OPTOTRAK, and surface EMGs were collected from Tibialis anterior, Gastrocnemius, Biceps femoris, and Rectus femoris muscles. The kinetic and kinematic data showed the characteristic features of the H&R paradigm: center of foot pressure initially remains unchanged after release as body momentum develops and then the Gastrocnemius and Biceps femoris muscles are activated and produce torques that move the center of pressure in front of the center of mass thus arresting the forward momentum of the body. The earliest EMG response was a simultaneous activation of Gastrocnemius and Biceps femoris at 50-60 ms in 1g. This initial reaction was significantly longer in 0g and shorter in 1.8g, but vision did not affect it. A second synergy involved Tibialis anterior and Rectus femoris, and it occurred later than 200 ms in 1g. The onset latency of the secondary reaction was significantly reduced in the presence of vision but was not affected by g level. The latencies of both synergies differed in the sequential 1g test periods. That is, latencies were longer in 1g in flight than 1g pre-flight and got longer still in 1g post-flight. The rate of body sway prior to EMG responses was not correlated with g level. This study shows that the H&R paradigm is feasible for postural evaluations in hypo-g and hyper-g force backgrounds as well as in 1g. G-related modulation of the earliest corrective EMG responses could be due to altered vestibulo-spinal modulation of muscle spindle gain, as a function of background force level. Later EMG responses to induced falls are under the control of vision. 2) Hold & Release During Galvanic Stimulation. To assess vestibular influences on recovery of balance during sudden falls we measured the postural responses of five healthy subjects to a hold and release perturbation coupled with galvanic vestibular stimulation (GVS). Two electrode pairs were located with the anterior electrode of each pair over the mastoid process and the posterior electrode over the trapezius muscle on the same side. Bilateral unipolar GVS was generated when a holding force against the sternum was released and the subject began moving forward, by individually driving left and right electrode pairs as cathode-anode or vice versa at 1 ma for 12 s or 2 ma for 6 s. We computed the frequency and damping parameters of a multi-link inverted pendulum model of the body which best fit the transient postural oscillations after release for each subject. These parameters did not differ significantly across conditions indicating that GVS did not modify the preset overall strategy of postural recovery. The intensity and polarity of GVS significantly biased both the postural lean during the oscillatory period and the resting postural stance after recovery, deviating them forward for cathodal-anodal stimulation and backward for anodal-cathodal. Because the GVS affected postural bias but not dynamics, it is likely that GVS either biased signals from otolith regular fibers or provided erroneous velocity signals leading to vestibulospinal compensations complementing the segmental stiffness and damping mechanism. The residuals of the multi-link fit, i.e., the frequency spectra of the actual body sway ripples about the modeled sway, were different across conditions, which also suggests that GVS affects the descending velocity and acceleration drive of the vestibulo-spinal pathway. This finding is consistent with recent theoretical analyses of the influence of GVS on the semicircular canals and otolith organs of the inner ear. 3) Response to Hold & Release in Patients with Bilateral Vestibular hypofunction. We measured the postural responses of six bilateral vestibular hypofunction patients and 19 control subjects to a hold and release test paradigm that induced a brief sudden forward pivot about the ankles. A linear four link inverted pendulum model was used to fit the resulting recovery of balance that was achieved without stepping. EMG latencies of the initial muscle responses to brake the forward tilt of the body did not differ across subject groups. Both subject groups exhibited linear time invariant behavior throughout the recovery period after release maintaining constant stiffness and damping. Both groups also exhibited significantly greater damping, energy dissipation, when light touch of the hand at mechanically non-supportive force levels or sight of the surroundings were permitted. The patients showed significantly less accurate control of the hip sway component than the normal subjects. The findings indicate that for both normal subjects and vestibular deficient subjects the pattern of response to be initiated at release is preset based on the body configuration and hold force prior to release and is executed as a unit.

4) The H&R paradigm reveals adaptation of long latency reflexes during exposure to a rotating artificial gravity environment. The aim of the present study was to quantify recovery from simulated falls in altered force backgrounds and to investigate the mechanisms of recovery, using the H&R paradigm. We tested 16 healthy men and women pre-, per-, and post- 10rpm rotation in the center of the Brandeis slow rotation room [2]. Subjects were divided into two groups: one group had an extra load equivalent to 25% of their body weight strapped to their waist, the other did not. Subjects voluntarily rocked back and forth about their ankles in 30s trials, ~3cm sway amplitude, for 3 pre-rotation, 20 per-rotation, and 10 post rotation trials. Loss of balance was elicited with the H&R perturbation paradigm in 3 pre-rotation trials and 3 trials immediately post-rotation. An OptoTrak system was used to measure postural sway and response to H&R.

Pre-rotation subjects successfully swayed fore and aft. During per-rotation trials they initially swayed in a deviated path, ~ 20º, reflecting the influence of Coriolis forces on the body. With additional trials, subjects swayed in a progressively more fore-aft pattern. Adaptation was fastest for the loaded subjects. Post-rotation subjects initially swayed in a mirror image path to that initially seen during rotation. The aftereffect was largest for the subjects with additional load. A comparison of pre- and post-rotation H&R trials indicated that the postural recovery response is not anti-parallel to the direction of the induced fall. Instead, it is in the same direction as the voluntary sway aftereffects, indicating a remapping of both voluntary sway and also of reflexive responses to postural perturbations. After 10 post-rotation sway trials, sway patterns returned to pre-rotation patterns.

Our earlier work showed the feasibility of using the H&R paradigm for studying postural responses in hypo- and hyper-g force environments. We show here that it is equally applicable to artificial gravity environments. Adaptation to artificial gravity results in remapping of postural reflexive responses as well as voluntary motor control.

The H&R paradigm can be used before, during and after spaceflight and after exposure to artificial gravity environments to quantify the capacity to recover from falls. It also aids in identifying factors contributing to post-flight postural ataxia.

Bibliography Type:	Description: (Last Updated: 03/31/2017)
Abstracts for Journals and Proceedings	Bakshi A, DiZio P, Lackner J. "Adaptation of postural control to Coriolos stimulation: role of effective body mass." Neuroscience 2007, Society for Neuroscience 37th annual meeting, San Diego, CA, November 3-7, 2007. Society for Neuroscience Abstracts, 2007. , Oct-2007
Abstracts for Journals and Proceedings	Bortolami SB, Chiovetto E, DiZio P, Lackner JR. "Postural responses to hold and release perturbation in parabolic flight." Neuroscience 2005. Society for Neuroscience's 35th annual meeting, Washington, DC, November 12-16, 2005. Society for Neuroscience Abstracts, 2005. , Oct-2005
Abstracts for Journals and Proceedings	Bortolami SB, Inglis JT, Castellani S, DiZio P, Lackner JR. "Monopolar galvanic stimulation during hold and release postural paradigm." Neuroscience 2004, Society for Neuroscience 34th annual meeting, San Diego, CA, October 23-27, 2004. Society for Neuroscience Abstracts, 2004. , Oct-2004
Abstracts for Journals and Proceedings	Bryan AS, Ventura J, Bortolami SB, DiZio P, Lackner J. "Localization of subjective vertical and head midline in altered gravitoinertial force environments." Neuroscience 2004, Society for Neuroscience 34th annual meeting, San Diego, CA, October 23-27, 2004. Society for Neuroscience Abstracts, 2004. , Oct-2004
Abstracts for Journals and Proceedings	Chiovetto E, DiZio P, Lackner JR. "Haptic stabilization of posture during quiet stance and rhythmic voluntary sway." Neuroscience 2007, Society for Neuroscience 37th annual meeting, San Diego, CA, November 3-7, 2007. Society for Neuroscience Abstracts, 2007. , Oct-2007
Abstracts for Journals and Proceedings	Frucht C, DiZio P, Lackner JR. "Discrete motor adaptations to simultaneous contacting and non-contacting loads." Neuroscience 2004, Society for Neuroscience 34th annual meeting, San Diego, CA, October 23-27, 2004. Society for Neuroscience Abstracts, 2004. , Oct-2004
Abstracts for Journals and Proceedings	Kim KH, Lackner JR, DiZio P. "Coordination of balance and reach movements." Neuroscience 2007, Society for Neuroscience 37th annual meeting, San Diego, CA, November 3-7, 2007. Society for Neuroscience Abstracts, 2007. , Oct-2007
Abstracts for Journals and Proceedings	Lackner JR, Ventura J, DiZio P. "Dynamic spatial orientation in altered gravitoinertial force environments." Neuroscience 2006, Society for Neuroscience 36th annual meeting, Atlanta, GA, October 14-18, 2006. Society for Neuroscience Abstracts, 2006. , Sep-2006
Abstracts for Journals and Proceedings	Pierobon A, Bortolami SB, Pigeon P, DiZio P, Lackner J. "Kinematics/kinetics correlations of arm motor control during Coriolis perturbations." Neuroscience 2005. Society for Neuroscience's 35th annual meeting, Washington, DC, November 12-16, 2005. Society for Neuroscience Abstracts, 2005. , Oct-2005
Abstracts for Journals and Proceedings	Piovesan D, DiZio P, Lackner JR. "Arm mechanical impedance measure using Impulse response." Neuroscience 2007, Society for Neuroscience 37th annual meeting, San Diego, CA, November 3-7, 2007. Society for Neuroscience Abstracts, 2007. , Oct-2007
Articles in Peer-reviewed Journals	Bortolami SB, Pierobon A, DiZio P, Lackner JR. "Localization of the subjective vertical during roll, pitch, and recumbent yaw body tilt." Experimental Brain Research. 2006 Aug;173(3):364-73. PubMed PMID: 16628401 , Aug-2006
Articles in Peer-reviewed Journals	Bortolami SB, Rocca S, Daros S, DiZio P, Lackner JR. "Mechanisms of human static spatial orientation." Exp Brain Res. 2006 Aug;173(3):374-88. PubMed PMID: 16628400 , Aug-2006
Articles in Peer-reviewed Journals	Bortolami SB, Pigeon P, DiZio P, Lackner JR. "Dynamics model for analyzing reaching movements during active and passive torso rotation." Exp Brain Res. 2008 Jun;187(4):525-34. PubMed PMID: 18330549 , Jun-2008

Articles in Peer-reviewed Journals	Bortolami SB, Pigeon P, DiZio P, Lackner JR. "Kinetic analysis of arm reaching movements during voluntary and passive rotation of the torso." Exp Brain Res. 2008 Jun;187(4):509-23. PubMed PMID: 18330550 , Jun-2008
Articles in Peer-reviewed Journals	Bryan AS, Bortolami SB, Ventura J, DiZio P, Lackner JR. "Influence of gravitoinertial force level on the subjective vertical during recumbent yaw axis body tilt." Exp Brain Res. 2007 Nov;183(3):389-97. PMID: 17703287 , Nov-2007
Articles in Peer-reviewed Journals	Hudson TE, Lackner JR, DiZio P. "Rapid adaptation of torso pointing movements to perturbations of the base of support." Experimental Brain Research 2005 Sep;165(3):283-93. PubMed PMID: 15942737 , Sep-2005
Articles in Peer-reviewed Journals	Lackner JR, DiZio P. "Angular displacement perception modulated by force background." Exp Brain Res. 2009 May 195(2):335-43. PubMed PMID: 19381620 , May-2009
Articles in Peer-reviewed Journals	Rabin E, DiZio P, Lackner JR. "Time course of haptic stabilization of posture." Experimental Brain Research 2006 Mar;170(1):122-6. PubMed PMID: 16501964 , Mar-2006
Articles in Peer-reviewed Journals	Rabin E, DiZio P, Ventura J, Lackner JR. "Influences of arm proprioception and degrees of freedom on postural control with light touch feedback." J Neurophysiol. 2008 Feb;99(2):595-604. PubMed PMID: 18032569 , Feb-2008
Articles in Peer-reviewed Journals	Wright WG, DiZio P, Lackner JR. "Perceived self-motion in two visual contexts: dissociable mechanisms underlie perceived self-motion." Journal of Vestibular Research 2006;16(1-2):23-8. PubMed PMID: 16917165 , Jan-2006
Articles in Peer-reviewed Journals	Wright WG, DiZio P, Lackner JR. "Vertical linear self-motion perception during visual and inertial motion: more than weighted summation of sensory inputs." J Vestib Res. 2005;15(4):185-95. PubMed PMID: 16286700 , Jan-2005
Articles in Peer-reviewed Journals	Lackner J, DiZio P. "Space motion sickness." Exp Brain Res. 2006 Nov;175(3):377-99. Review. PubMed PMID: 17021896 , Nov-2006
Books/Book Chapters	Lackner JR, DiZio P. "Multisensory influences on orientation and movement control." in "The handbook of multisensory processes." Ed. G.A. Calvert, C. Spence, B.E. Stein. Cambridge, Mass. : MIT Press, c2004. p. 409-423., Aug-2004
Books/Book Chapters	Lackner J, DiZio P. "Proprioception: Effect of gravity." in "Encyclopedia of Neuroscience." Ed. M.D. Binder, N. Hirokawa, U. Windhorst. New York : Springer, 2008., Dec-2008